Topics

Section 1.2: σ-algebra

- Problems with the old definitions of probability and a discussion of a new definition of probability.
- Introducing σ-algebras for probability spaces.
- Introducing measurable sets and measurable spaces.

Section 1.3: Measures

- Introducing measures.
- Examining the Dirac measure as an example.

Section 1.4: Monotone continuity of measures

- Monotonicity of measures.
- Monotone continuity of measures.
- The union bound.

Section 1.6 discusses discrete probability spaces and probability mass functions (pmf). Since these have been explained in detail in the course, they are not addressed here.

Section 1.2

Old definitions of probability

Classical probability

$$P(A) = \frac{|A|}{|S|}.$$

There are two problems with this definition:

1. It is restricted to a discrete space.
2. It is not applicable to infinite cases; for example, calculating the probability of obtaining an even number when a natural number is selected at random.

The relative-frequency definition of probability

$$P(A) = \lim_{n\to\infty} \frac{\text{number of occurrences of event } A \text{ in } n \text{ trials}}{n}.$$

The point in this definition is that we have assumed that an event and a probability already exist, even though we are only now about to define them. However, the definition above takes this concept for granted. Thus, before defining probability, we have assumed it. We have encountered a logical flaw. (In fact, relative frequency is better regarded as a result of probability theory, rather than as its definition.)

Unlike the previous definition, this definition can also be used for infinite spaces. Its major problem, however, is that it does not have the property of countable additivity; that is,

$$\forall i, j \in \mathbb{N}, \quad A_i \cap A_j = \varnothing \quad \not\Rightarrow \quad P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

For example, suppose that we want to select a natural number in this way. Clearly, the probability of "selecting a natural number" is 1. But under the relative-frequency definition, the probability of selecting any particular natural number is zero. In other words, if A_i is the event that the number i is selected, then

$$\forall i \in \mathbb{N}: \quad P(A_i) = 0.$$

Consequently,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = 1, \qquad \sum_{i=1}^{\infty} P(A_i) = 0 \quad \longrightarrow \quad \text{countable additivity does not hold.}$$

Yet we know that countable additivity is one of the key properties in the definition of probability. It enables us to assign a probability to events that are countable unions of events. Many probability concepts, such as expectation and other properties, are defined on this basis.

The geometric definition of probability

$$P(A) = \frac{\text{length / area / volume of event } A}{\text{total length / area / volume}}.$$

Unlike the previous two definitions, the problem with this definition is not obvious.

This notion of probability makes an assumption: a length (and hence a probability) can be assigned to every set. Below, we show that a set can be found for which length cannot be defined. Consequently, $P(A)$ has no value in the sense described above, and the definition above cannot be a correct definition of probability.

Here we examine a strange case: a set, V, to which no length can be assigned.

Consider the interval $I = [0, 1]$. We regard $x, y \in I$ as equivalent whenever their difference is a rational number:

$$x \sim y \quad \Longleftrightarrow \quad x - y \in \mathbb{Q}.$$

For every $x \in I$, there is exactly one equivalence class in the following sense:

$$[x] = \{w \in I \mid w - x \in \mathbb{Q}\}.$$

(Clearly, for different real numbers, these classes are either identical or completely disjoint. From now on, we consider distinct classes.)

- I is the union of disjoint equivalence classes. Each $x \in I$ belongs to exactly one class.
- Each class is countable: it is a translation of the rational numbers.
- There are uncountably many classes. Suppose that there were countably many. Each class contains countably many elements, so their union would also be countable: a countable union of countable sets is countable. But their union is I, which is uncountable. This is a contradiction.

To proceed with the argument, we need to introduce an axiom.

Axiom of choice. Given an uncountable family of sets, we can choose one element from each set.

Note. This is an existence axiom. It does not provide a method for making the choices or say how to make them.

Now select one representative from each equivalence class. The axiom of choice guarantees that a representative can be chosen from each of the uncountably many classes. We call the set of these representatives a **Vitali set**, and denote it by V.

No two members of a Vitali set have a rational difference: the members belong to different equivalence classes.

We shall have

$$\forall q \in [-1, 1], \qquad V + q = \{v + q \mid v \in V\}.$$

(Each element of the set is shifted by q.)

The sets $V + q$ are pairwise disjoint for different values of q.

Proof. By contradiction. Suppose that, for two different values $q_1 \neq q_2$, the sets $V + q_1$ and $V + q_2$ have a common element. Then

$$v_1 + q_1 = v_2 + q_2, \qquad v_1, v_2 \in V.$$

The elements v_1, v_2 belong to different equivalence classes, whereas

$$v_1 - v_2 = q_2 - q_1 \quad \longrightarrow \quad v_1 - v_2 \in \mathbb{Q},$$

which is a contradiction.

The union of $V + q$ for $q \in \mathbb{Q} \cap [-1, 1]$ reconstructs the set I, since it reconstructs the equivalence class associated with each representative. Previously, we had the union of the distinct equivalence classes. Thus,

$$[0, 1] \subset \bigcup_{q \in \mathbb{Q} \cap [-1,1]} (V + q).$$

Annotation to this inclusion. One representative is selected from each class and added to all rational numbers in the interval $[-1, 1]$, so some points in the resulting union lie outside I. For example, suppose that 0 is the representative of an equivalence class. Then

$$-1 \in \bigcup_{q \in \mathbb{Q} \cap [-1,1]} (V + q).$$

This is why $\subset$ has been used, rather than $=$. The measure of these points outside the interval is zero.

To understand the relation above, we know that each real number in the interval has a rational difference from one of the representatives, that is, from a member of V. (Their difference is the amount of the rational translation.) The union above shifts every representative by all rational numbers in $[-1, 1]$.

Here we define the function $\ell(\cdot)$ as a length function with the following properties:

0) $\ell(V) \geq 0$.

1) $\ell(V + q) = \ell(V)$: length does not change under translation.

2) ℓ is countably additive.

Thus we shall have

$$[0, 1] \simeq \bigcup_{q \in \mathbb{Q} \cap [-1,1]} (V + q) \quad \Rightarrow \quad \ell([0, 1]) = \ell\left(\bigcup_{q \in \mathbb{Q} \cap [-1,1]} (V + q) \right). \tag{1}$$

Using property 2 in (1), and then property 1,

$$1 = \sum_{q \in \mathbb{Q} \cap [-1,1]} \ell(V + q) \quad \Longrightarrow \quad \boxed{1 = \sum_{q \in \mathbb{Q} \cap [-1,1]} \ell(V)}.$$

Thus, the length of V is added to itself infinitely, countably many times:

$$\begin{cases} \ell(V) = 0 & \Rightarrow \quad 1 = 0 \quad \text{(contradiction)}, \\ \ell(V) > 0 & \Rightarrow \quad 1 = \infty \quad \text{(contradiction)}. \end{cases}$$

No length can therefore be assigned to the set V. In other words, we can find a set for which the geometric definition of probability encounters this problem.

A definition of probability that addresses the problems above

A triple $(S, \mathcal{S}, P)$ is called a probability space, where

- S is the sample space.
- $\mathcal{S}$ (a σ-algebra) consists of subsets to which a probability can be assigned. We define a σ-algebra precisely below.
- P is a function from $\mathcal{S}$ to $[0, 1]$ satisfying certain conditions. It suffices for P to be a probability measure. Measures and their properties are introduced precisely in the next section.

σ-algebra ($\mathcal{S}$)

A family of subsets of the sample space S that contains $\varnothing$ and S and has the following properties:

$$\begin{aligned}
&\text{i)} \quad A \in \mathcal{S} \quad \Rightarrow \quad A^c \in \mathcal{S} && \text{(closed under complementation)},\\
&\text{ii)} \quad A_1, A_2, \ldots \in \mathcal{S} \quad \Rightarrow \quad A_1 \cup A_2 \cup \cdots \in \mathcal{S} && \text{(closed under countable unions)},\\
&\text{iii)} \quad A_1, A_2, \ldots \in \mathcal{S} \quad \Rightarrow \quad A_1 \cap A_2 \cap \cdots \in \mathcal{S}.
\end{aligned}$$

(The properties are not independent. The third can be derived from the first and second using De Morgan's law.)

The intuition behind introducing a σ-algebra is to resolve the problems of the classical definitions. By restricting the definition of probability to the "right" sets, it becomes possible to define probability without difficulty.

- The members of a σ-algebra are called **measurable sets**.
- The pair $(S, \mathcal{S})$ is called a **measurable space**.

For example, the following are σ-algebras:

$$\mathcal{S} = 2^S \qquad \text{(a countable probability space)},$$

$$\mathcal{S} = \{\varnothing, S\} \qquad \text{(a trivial } \sigma\text{-algebra)}.$$

Section 1.3: The concept of measure and probability measure

In the preceding section, we discussed the concept and definition of a σ-algebra. We also saw that, in the precise definition of probability, it suffices to assign numbers to the members of a σ-algebra. Hence, it is enough to have a measure. We introduce measures precisely below.

On a measurable space $(S, \mathcal{S})$, a function $\mu : \mathcal{S} \to [0, \infty]$ is called a **measure** if it satisfies the following two conditions:

$$\begin{aligned}
&\text{i)} \quad \mu(\varnothing) = 0,\\
&\text{ii)} \quad \mu\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mu(A_n), \qquad A_i \in \mathcal{S},\\
&\qquad\qquad\qquad\qquad \forall i, j \in \mathbb{N} : \; i \neq j \;\Rightarrow\; A_i \cap A_j = \varnothing.
\end{aligned}$$

The second condition is **countable additivity**.

The intuition behind the second condition is the additivity of length, area, and volume for disjoint sets. Its purpose is to extend the properties of countable unions (and intersections) in the definition of probability.

After introducing measures, we return to the definition of probability. A measure is a method for assigning a number to events to which we want to assign a probability.

If $\mu(S) < \infty$ (the measure of the sample space is finite), then μ is a **finite measure**.

If $\mu(S) = 1$ (the measure of the sample space equals one), then μ is a **probability measure**.

In fact, we obtain a probability measure by normalizing a finite measure. It is now clear why we said that P suffices in the previous precise definition of probability: it is in fact a probability measure and has the properties of a measure.

We now examine an example and show that it is a measure: the Dirac measure.

Dirac measure

At a point $x \in S$ in a measurable space $(S, \mathcal{S})$, the Dirac measure is a function $\delta_x : \mathcal{S} \to [0, \infty]$ defined by

$$\delta_x(A) = \begin{cases} 1, & x \in A, \\ 0, & \text{otherwise.} \end{cases}$$

To show that δ_x is a measure, we must show that it has both properties of a measure. Clearly, $\delta_x(\varnothing) = 0$.

To check the second property, take $A_1, A_2, \ldots \in \mathcal{S}$ to be pairwise disjoint. There are two cases for x.

i) $x \notin \bigcup_{n\geq 1} A_n$.

By the assumption in this case,

$$\delta_x\left(\bigcup_{n\geq 1} A_n\right) = 0.$$

Moreover, because the sets A_i are disjoint,

$$\forall n \in \mathbb{N}, \quad \delta_x(A_n) = 0 \quad \Rightarrow \quad \sum_{n\geq 1} \delta_x(A_n) = 0.$$

Thus, in this case,

$$\delta_x\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \delta_x(A_n).$$

ii) $x \in \bigcup_{n\geq 1} A_n$.

By the assumption in this case,

$$\delta_x\left(\bigcup_{n\geq 1} A_n\right) = 1.$$

Because the sets A_i are pairwise disjoint,

$$\exists k \in \mathbb{N}, \quad \delta_x(A_k) = 1, \qquad \forall n \in \mathbb{N},\ n \neq k \quad \Rightarrow \quad \delta_x(A_n) = 0.$$

Therefore,

$$\sum_{n\geq 1} \delta_x(A_n) = 1.$$

We see that both conditions for a measure hold. Thus δ_x is a measure.

Also, by the definition of δ_x, we have $\delta_x(S) = 1$. Hence the Dirac measure is a probability measure.

Section 1.4: Properties of measures

In this section, we derive several important results from countable additivity, that is, from the definition of a measure.

1. Monotonicity

Every measure μ is monotone. This means that, for any measurable sets A, B,

$$A \subset B \quad \Rightarrow \quad \mu(A) \leq \mu(B).$$

The intuition behind this property is clear: if A is entirely contained in B, it cannot have a measure greater than that of B.

Proof.

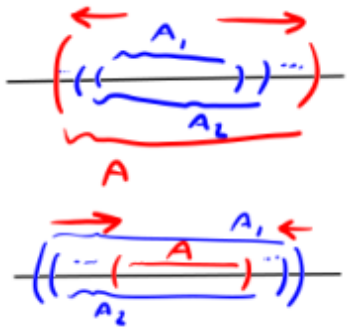
$$A \subset B \quad \Rightarrow \quad B = \underbrace{A \cup (A^c \cap B)}_{\text{disjoint}}.$$

Using closure under unions, intersections, and complements,

$$\mu(B) = \mu(A) + \underbrace{\mu(A^c \cap B)}_{\mu(\cdot)\geq 0} \quad \longrightarrow \quad \mu(B) \geq \mu(A).$$

Monotone limits of sets

$$\text{i)} \quad A_n \uparrow A \quad \text{means} \quad A_1 \subset A_2 \subset \cdots, \qquad \bigcup_{n\geq 1} A_n = A,$$
$$\text{ii)} \quad A_n \downarrow A \quad \text{means} \quad A_1 \supset A_2 \supset \cdots, \qquad \bigcap_{n\geq 1} A_n = A.$$



Examples:

$$A_n = \left[0, 1 - \frac{1}{n}\right], \quad A = [0,1] \quad \longrightarrow \quad A_n \uparrow A,$$
$$A_n = \left[-\frac{1}{n}, 1\right], \quad A = [0,1] \quad \longrightarrow \quad A_n \downarrow A.$$

Similar notation is used for monotone continuity in $[0, \infty]$. For example,

$$x_n \uparrow x \quad \text{means} \quad x_1 \leq x_2 \leq \cdots, \qquad \lim_{n\to\infty} x_n = x.$$

2. Monotone continuity

Every measure μ is continuous under monotone limits. This means that, for every measurable set,

$$\text{i)} \quad A_n \uparrow A \quad \Rightarrow \quad \mu(A_n) \uparrow \mu(A),$$
$$\text{ii)} \quad A_n \downarrow A \;\&\; A_1 < \infty \quad \Rightarrow \quad \mu(A_n) \downarrow \mu(A).$$

Intuition. This property says that if sets converge monotonically, by intersection or union, to a set, then their measures also converge monotonically to the measure of that set.

Proof. The proof of this property is not essential for the continuation of the discussion, but it is included for a better understanding of the subject.

Proof of i). Take measurable sets such that $A_n \uparrow A$. Define the sets B_i by

$$B_1 = A_1, \qquad B_j = A_j \cap A_{j-1}^c \quad (j \neq 1).$$

Then

$$A_n = B_1 \cup \cdots \cup B_n.$$

The sets B_i are disjoint and $B_i \in \mathcal{S}$. Hence, by countable additivity,

$$\mu(A_n) = \sum_{j=1}^{n} \mu(B_j).$$

By the definition of a measure, each $\mu(B_j)$ is nonnegative. Consequently, as $n \to \infty$, the expression on the right converges monotonically to $\sum_{k=1}^{\infty} \mu(B_k)$.

$$\mu(A_n) \uparrow \sum_{k=1}^{\infty} \mu(B_k). \tag{1}$$

Moreover, by the definition it is clear that

$$\bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} B_k.$$

Thus, by countable additivity, since the B_i are disjoint,

$$\mu(A) = \mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \mu\left(\bigcup_{k=1}^{\infty} B_k\right) = \sum_{k=1}^{\infty} \mu(B_k).$$

Using (1), the claim is proved.

Proof of ii). Take measurable sets such that $A_n \downarrow A$ and $\mu(A_1) < \infty$, and define B_i and B by

$$B_i = A_1 \cap A_i^c, \qquad B = A_1 \cap A^c.$$

Then

$$A_1 = \underbrace{A_i \cup B_i}_{\text{disjoint}} = A \cup B.$$

By countable additivity,

$$\mu(A_1) = \mu(A_i) + \mu(B_i) = \mu(A) + \mu(B).$$

Since $\mu(A_1) < \infty$,

$$\mu(A_i) = \mu(A_1) - \mu(B_i), \tag{1}$$
$$\mu(A) = \mu(A_1) - \mu(B). \tag{2}$$

It follows from the definition that $B_n \uparrow B$, so by part i),

$$\mu(B_n) \uparrow \mu(B).$$

Consequently, using (1),

$$\begin{aligned} \lim_{n\to\infty} \mu(A_n) &= \lim_{n\to\infty} (\mu(A_1) - \mu(B_n)) \\ &= \mu(A_1) - \lim_{n\to\infty} \mu(B_n) \\ &= \mu(A_1) - \mu(B) = \mu(A), \end{aligned}$$

where the last equality follows from (2).

Importance of this theorem. If we have a complicated set that cannot easily be measured, we can "approximate" it by simple sets and use this theorem to calculate its measure. In other words, if we have a sequence of simple sets converging to the set in question, their measures will also converge to the measure of that set.

This has many applications in random functions and probability theory: it makes it possible to calculate the probability of a limiting event. Calculating the probability of "happening" and "never happening," for example, depends on this property.

3. The union bound

For every measure μ, we can write

$$\mu\left(\bigcup_{i\geq 1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

for every finite or countable family of measurable sets.

Intuition. The sets may overlap. When their measures are added, these parts are counted several times. The measure of the union is therefore at most the sum of their measures; equality occurs when all the sets are disjoint.

Proof. For two sets,

$$\mu(A \cup B) = \mu(A) + \mu(B \cap A^c) \leq \mu(A) + \mu(B),$$

where the inequality follows from monotonicity.

For the finite case, we can extend the result above by induction:

$$\mu\left(\bigcup_{k=1}^{n} A_k\right) \leq \sum_{k=1}^{n} \mu(A_k).$$

For the countable case, take

$$C_n = \bigcup_{k=1}^{n} A_k, \qquad C = \bigcup_{k=1}^{\infty} A_k.$$

Then $C_n \uparrow C$. By monotone continuity of measures from the preceding section,

$$\mu(C_n) \uparrow \mu(C).$$

On the other hand, the finite-case result gives

$$\mu(C_n) = \mu\left(\bigcup_{k=1}^{n} A_k\right) \leq \sum_{k=1}^{n} \mu(A_k),$$

and

$$\sum_{k=1}^{n} \mu(A_k) \uparrow \sum_{k=1}^{\infty} \mu(A_k).$$

(It does not matter here whether this expression is infinite; the measures can still be added.)

For every n,

$$\mu(C_n) \leq \underbrace{\sum_{k=1}^{\infty} \mu(A_k)}_{\text{constant}}.$$

Hence,

$$\lim_{n\to\infty} \mu(C_n) \leq \sum_{k=1}^{\infty} \mu(A_k),$$

so

$$\mu(C) = \mu\left(\bigcup_{k=1}^{\infty} A_k\right) \leq \sum_{k=1}^{\infty} \mu(A_k).$$

The claim is thus proved for the countably infinite case as well.

The importance and application of this theorem in examining the existence of coupling and finding bounds on the distance between measures is fundamental.